What the correlations between neurons say about each neuron

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Draft v0.1, September 11, 2026

Abstract

Record many neurons while they watch the same stimulus, then discard the stimulus. What remains is how the neurons relate to each other: a correlation matrix. We ask whether that matrix alone says what each neuron represents. A transformer that sees only the correlation matrix of 512 sampled neurons, with no labels and no reference population in its input, assigns preferred orientation to neurons of mouse visual cortex at 0.38 accuracy (8 classes, majority class 0.26) and receptive-field position at $R^2 = 0.23$, from in-vivo responses in the MICrONS volume; on the digital-twin substrate it reaches 0.42 and 0.48. The readout is possible because the relations between orientation classes are close to circulant, which fixes the class structure up to rotation and reflection, and a small anisotropy pins the absolute frame. Across 33 mice of the Allen Brain Observatory, orientation transfers to unseen animals only when a sampled population mixes animals, so that its matrix contains between-animal correlations; populations drawn from one animal sit at the baseline whether or not the animal was seen in training. For receptive fields, what crosses animals is where each animal's imaged patch looks on the screen, and not the layout of neurons within it. Capacity helps where labels are plentiful and stops helping where they are scarce.

1 Introduction

What a neuron in visual cortex represents is established by relating its activity to the stimulus: a neuron prefers vertical gratings, or responds to the upper left of the screen. This paper asks whether the same facts are contained in a description that mentions no stimulus at all, the correlations between the neuron and the other neurons recorded with it. Take a population that watched the same movie, compute the correlation of every pair of neurons, and discard the movie. Does the resulting matrix determine which neuron prefers which orientation, and where each neuron's receptive field lies?

The idea that content can be read from relations is old. Shepard and Chipman proposed that internal representations need not resemble what they stand for as long as the similarities among them mirror the similarities among the things represented, a second-order isomorphism [1]; Edelman made the case that representation is representation of similarities [2]. Representational similarity analysis turned the idea into a method that compares brains, models and species through their similarity structures rather than their coordinates [3, 4], hyperalignment builds a shared representational space across subjects from those structures [5], and the same principle underlies current work on aligning representations across networks and individuals [6, 7]. In neuroscience, the correlation structure of spontaneous activity was shown to carry the orientation map of visual cortex without any stimulus [8, 9] and, more generally, to match the statistics of evoked activity [10]; functionally similar neurons connect preferentially [11, 12]; and population dynamics are preserved across animals performing the same behaviour [13]. Structuralist accounts of experience make the general claim that what a state represents, or how it feels, is fixed by its position in a space of relations [14], with Kleiner and Ludwig [15] giving a formal definition of

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such structures; Lässig [16] formulates the corresponding question for neural networks and gives the decoding framework we use here.

The spontaneous-activity results are the closest precedent, and they set the premise: a neuron's correlations with the population contain its tuning even in the dark. Tsodyks et al. [8] showed it for single neurons by averaging the imaged population activity at the times a neuron fired and finding the evoked map of its preferred orientation. Reading the tuning out of that pattern, however, required the evoked map, a stimulus-labelled reference. The same holds for representational similarity analysis and hyperalignment, whose structures are over conditions or subjects and whose interpretation goes through a known stimulus set. Our question is one level down and one step further: whether the relations among *neurons*, with the stimulus that produced them discarded and no labelled neuron or map to compare with, fix the content of each neuron, and whether that content is readable at the scale of thousands of individually identified neurons and across animals. In practice this is often the situation one is in: recordings under natural behaviour have no controlled stimulus, and neurons from different sessions or animals share no stimulus-aligned coordinate system, but they do share the structure of their correlations. In principle, the answer says how much of a neuron's content is fixed by its position in the network of relations among neurons.

We answer it with a decoder. The input is the correlation matrix of a sampled population of a few hundred neurons, one row per neuron, and nothing else: no labels and no reference neurons. The output is a preferred orientation and a receptive-field position for every neuron in the population. The decoder is trained on populations whose labels are known and scored on neurons it never saw. We do this at two scales. Within one brain, using the MICrONS release [12, 17], 12,894 neurons from a cubic millimetre of mouse visual cortex with in-vivo responses to a shared movie, a fitted digital-twin model [18], and labels for orientation and receptive field (Fig. 1): the decoder trained on half the neurons reads the other half. Across brains, using the Allen Brain Observatory [19], 33 mice recorded under identical stimuli: the decoder trained on some animals reads neurons of animals it never saw.

The results come with an explanation. Averaging the correlations by orientation class shows that the relations between classes are almost circulant, which fixes the classes only up to a rotation or reflection of the orientation circle; a small, consistent departure from that symmetry is what pins the absolute frame and makes the readout possible.

2 Methods

2.1 Data

MICrONS. The release of Ding et al. [12] contains 12,894 neurons co-registered to the electron-microscopy volume, from 13 two-photon scans across V1, RL, AL and LM. Each neuron has a trial-averaged in-vivo response to the oracle natural-movie clips (120 bins), shown in every scan; a digital-twin response to a shared movie (4,999 bins, which we compress to 512 principal components; the Gram is preserved to $r > 0.999$); an in-vivo preferred orientation with a global orientation selectivity index (gOSI); and a receptive-field centre from the twin (spike-triggered-average fit, screen coordinates in $[-1, 1]$). We use orientation for the 5,287 neurons with gOSI ≥ 0.25, binned into 8 classes, and receptive-field position for the 11,326 neurons whose twin test correlation is ≥ 0.2. Same-scan pairs correlate about 50% more strongly than cross-scan pairs at matched receptive-field distance; the content of the correlations is the same across scans, and all populations mix scans.

Allen. From Visual Coding 2P [19] we take the 36 VISp excitatory containers with at least 150 orientation-labelled cells: 33 mice, 9,281 cells. Relations come from session A, either responses to natural movies shown to every mouse (4,500 bins) or the 40 blank-subtracted condition means

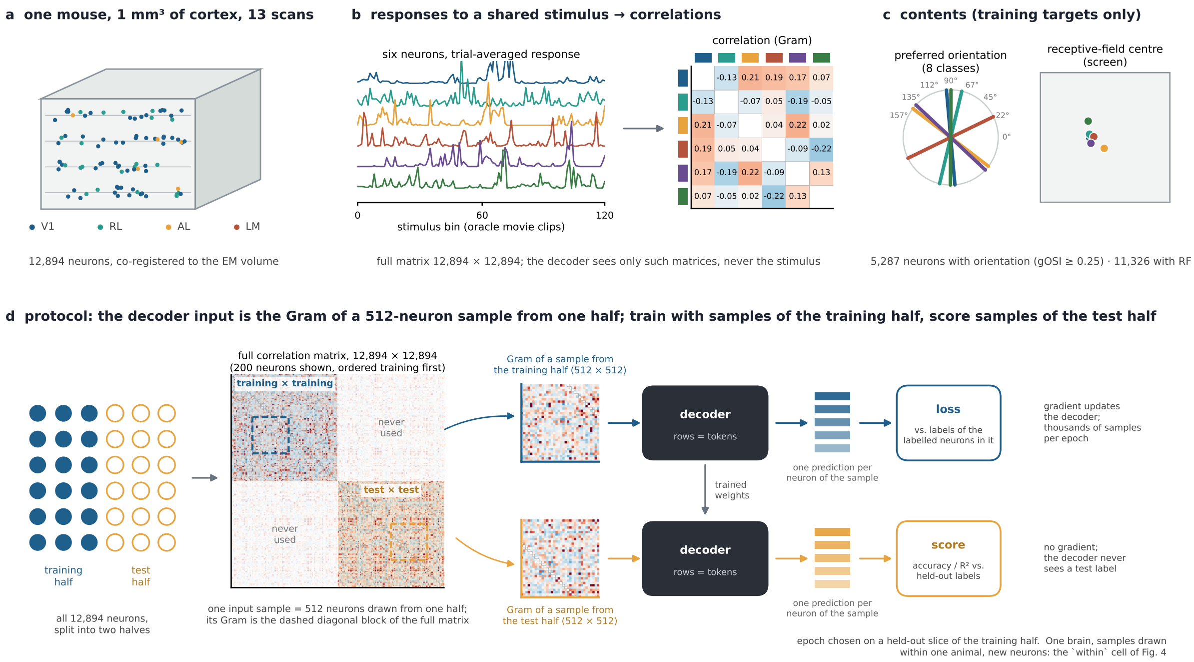

Figure 1: How the MICrONS task is built. (a) One mouse, 13 two-photon scans of a cubic millimetre of visual cortex (areas V1, RL, AL, LM), 12,894 neurons co-registered to the electron-microscopy volume. (b) The relational substrate. Each neuron is its trial-averaged response vector to a stimulus all neurons saw (in vivo: 120 bins of the oracle natural-movie clips; digital twin: 4,999 bins compressed to 512 principal components); shown are six real neurons and their correlation matrix. The oracle clips were shown in all 13 scans, so the matrix is defined across scans. The decoder only ever sees such correlations, never the stimulus. (c) The contents of the same six neurons: preferred orientation (in vivo, 8 classes, 5,287 neurons with gOSI ≥ 0.25) and receptive-field centre (digital-twin fit, 11,326 neurons). These are training targets only. (d) The protocol, with two terms kept apart. The *halves*: all neurons are split at random into a training half and a test half, and the full $12{,}894 \times 12{,}894$ correlation matrix, ordered training-first (a real 200-neuron excerpt is shown), is used only in its training×training and test×test blocks. A *sample*: 512 neurons drawn at random from one half; its Gram is the corresponding diagonal sub-block (dashed), and that 512×512 matrix is the decoder's entire input, one row per neuron, one prediction per row. Training draws thousands of samples per epoch from the training half; their predictions meet the labels of the labelled neurons in the sample in a loss that updates the decoder. Scoring draws samples from the test half and compares predictions with held-out labels; no gradient. Labels never enter the input on either path, and the halves share no neuron. The epoch is chosen on a held-out slice of the training half.

of drifting gratings (8 directions × 5 temporal frequencies). Orientation labels come from static gratings in session B (6 classes, gOSI ≥ 0.25, $p < 0.05$), so labels and relations use different stimuli. Receptive-field centres come from locally sparse noise in session C (1,504 cells with a significant fit, 23 to 112 per mouse).

2.2 Relations

Each neuron's response vector is centred and normalised; the relation between two neurons is the cosine of their vectors, a signal correlation. For a population of n neurons the Gram is the $n \times n$ matrix of these values, standardised by the global off-diagonal mean and standard deviation, with the diagonal zeroed.

2.3 Decoder

Each row of a population's Gram, together with three row statistics, is projected to a token; L pre-norm transformer blocks with H-head attention update the tokens; a linear head reads out an orientation class or an (x, y) pair per token. The loss is cross-entropy or mean squared error on the labelled tokens only; unlabelled neurons stay in the population and receive no loss. The standard configuration is width 512, 8 layers, 17M parameters, populations of 512 neurons (MICrONS) or 256 (Allen). Smaller (2.2M) and larger (57M) models are reported in the appendix. Training draws thousands of random populations per epoch. For Allen orientation, each training population's Gram is recomputed from a random half of the 40 conditions; test Grams use all conditions.

2.4 Protocol

Split. MICrONS neurons are halved at random. Training populations are sampled inside the training half and test populations inside the test half, so no test neuron ever appears in a training population (Fig. 1d). A slice of the training half is held out to choose the epoch; the test half never influences selection.

Regimes (Allen). Two independent choices define a run (Fig. 4). The split decides where test neurons come from: the training animals, each halved into training and test neurons, or held-out animals. The population decides what a sample is: neurons drawn from one animal, so that the Gram is a within-circuit matrix, or from several animals, so that most entries are between-animal correlations. We call the four cells `within`, `pooledwithin`, `cross` and `pooledcross`. Held-out regimes select the epoch on further held-out mice. Every held-out cell is run on three mouse splits and every training-animal cell on three neuron splits.

Baselines. The majority class (0.255 in MICrONS, 0.19 to 0.20 on Allen test mice) and $R^2 = 0$.

2.5 Symmetry analysis

To expose the symmetry of the relations we average the Gram into a $K \times K$ class-Gram, the mean relation between neurons of two orientation classes, project it onto its circulant part, and ask which relabellings of the classes a held-out half's class-Gram is compatible with (all $K!$ relabellings, Frobenius distance). A relabelling that differs from the truth by a rotation or a reflection of the orientation circle is an element of the dihedral group D_K; the analysis reports whether the truth is singled out absolutely or only modulo D_K.

Table 1: MICrONS, label-free per-neuron decoding on the test half; 17M decoder, 512-neuron populations, mean ± s.d. over 3 seeds.

content	substrate	score	baseline
orientation (8 classes, accuracy)	in vivo	0.380 ± 0.006	0.255
orientation (8 classes, accuracy)	digital twin	0.420 ± 0.003	0.255
receptive field (R^2)	in vivo	0.234 ± 0.007	0
receptive field (R^2)	digital twin	0.480 ± 0.010	0

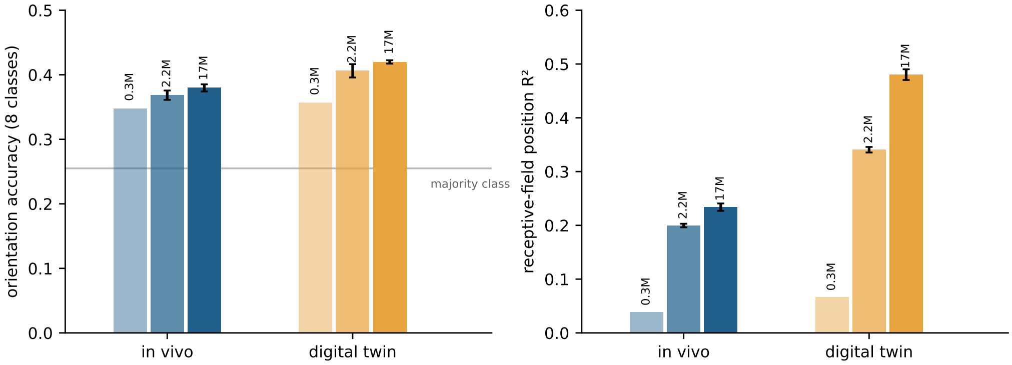

Figure 2: MICrONS: per-neuron content decoded from the population correlation matrix alone. Accuracy (left, preferred orientation, 8 classes) and R^2 (right, receptive-field centre) on held-out neurons of the same animal, for decoders of 0.3M, 2.2M and 17M parameters; error bars are the s.d. over 3 seeds. The decoder sees only the standardised correlation matrix of 512 sampled neurons. Grey line: majority class.

3 Results

3.1 Per-neuron content from the correlation matrix alone

On MICrONS the decoder reads orientation and receptive-field position from the Gram of 512 neurons of the test half (Fig. 2, Table 1). The twin substrate is cleaner than the in-vivo one, since the twin removes trial noise, and receptive fields carry more than orientation. The predictions are in the true frame: orientation accuracy modulo D_8 equals plain accuracy in every run, so the decoder places each neuron at its absolute orientation rather than recovering the class structure up to a rotation.

3.2 Why the absolute frame is recoverable

The class-Gram of orientation is close to circulant (Fig. 3): neighbouring orientations correlate, orthogonal ones anti-correlate, and the pattern repeats around the circle. The circulant part holds 71% of the variance in vivo and 81% in the twin. A circulant class-Gram is invariant under every rotation and reflection of the orientation circle, so it cannot single out which class is 0°: after projecting onto the circulant part, a held-out half's class-Gram is compatible with all 16 elements of D_8 and with nothing else (absolute matching at chance, matching modulo D_8 at 1.0). The raw class-Gram singles out the true labelling. The difference is the residual anisotropy, a cardinal bias whose strength differs between areas, and that is what pins the frame. A label-free decoder that outputs absolute orientation must use it; the results of the previous section show that it does.

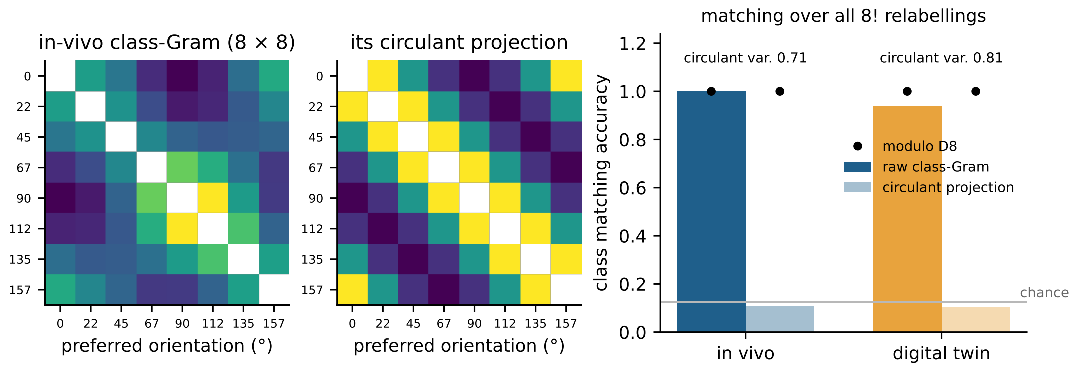

Figure 3: What fixes the absolute orientation frame. Left: mean correlation between orientation classes in vivo (diagonal masked). Middle: its circulant projection. Right: matching a held-out half's class-Gram to the reference over all 8! relabellings. With the raw matrix the true labelling wins. After the circulant projection, absolute accuracy collapses to chance while accuracy modulo D_8 (dots) stays at 1.0.

Table 2: Allen, orientation accuracy (6 classes) in the four regimes, three splits per cell (of neurons within each animal for the top row, of animals for the bottom row); the `pooledcross` split-0 value is the mean of 3 seeds. Majority-class baseline 0.19 to 0.20.

test neurons from	single-animal populations	mixed populations
the training animals	0.210 / 0.216 / 0.217	0.281 / 0.283 / 0.283
held-out animals	0.199 / 0.197 / 0.187	0.293 / 0.236 / 0.264

3.3 Orientation transfers between animals through mixed populations

With natural-movie relations, orientation is nearly absent from Allen correlations: same-orientation pairs correlate 0.005 more than orthogonal pairs, against 0.09 in MICrONS, and every test is at chance. With grating relations the signal is present (0.05 within a mouse, 0.044 across mice), and the decoder's four cells separate (Fig. 5, Table 2). Single-animal populations sit at the baseline whether or not the animal was seen in training. Mixed populations are above it in both rows, at the same level. Transfer to a new brain costs nothing; what matters is whether the Gram contains between-animal correlations. Those exist only because all mice saw the same 40 conditions, and they measure how similar two tuning profiles are. The decoder still never sees which condition is which, so the absolute frame must again come from the population's relational structure. In this dataset a neuron's orientation is readable from where it sits in a correlation geometry that spans animals; it is not readable from its relations inside its own circuit.

3.4 Receptive fields: what crosses animals is the animal's screen position

Correlation falls with receptive-field distance within a mouse (0.127 at $< 5°$ to 0.068 beyond $40°$) and across mice (0.049 to 0.011). Decoding absolute screen coordinates on held-out mice gives R^2 that swings with the split, from 0.29 to -0.45 for single-animal populations and from 0.21 to -0.25 for mixed ones. Decomposing the saved predictions explains the swing (Fig. 6). The correlation between predicted and true mean position of each held-out mouse, pooled over 27 mouse-level predictions, is 0.36 to 0.71 (permutation $p \leq 0.03$). The correlation between predicted and true position of a neuron relative to its mouse-mates is 0.0 to 0.1 over 1,259 cells. The decoder recovers where an animal's imaged patch looks, a property of the population, and not which neuron within it looks where. R^2 on absolute coordinates is then dominated by whether the between-mouse spread of a particular test set is reproduced at the right scale.

The within-mouse layout is present in the correlations, since same-mouse correlation falls with

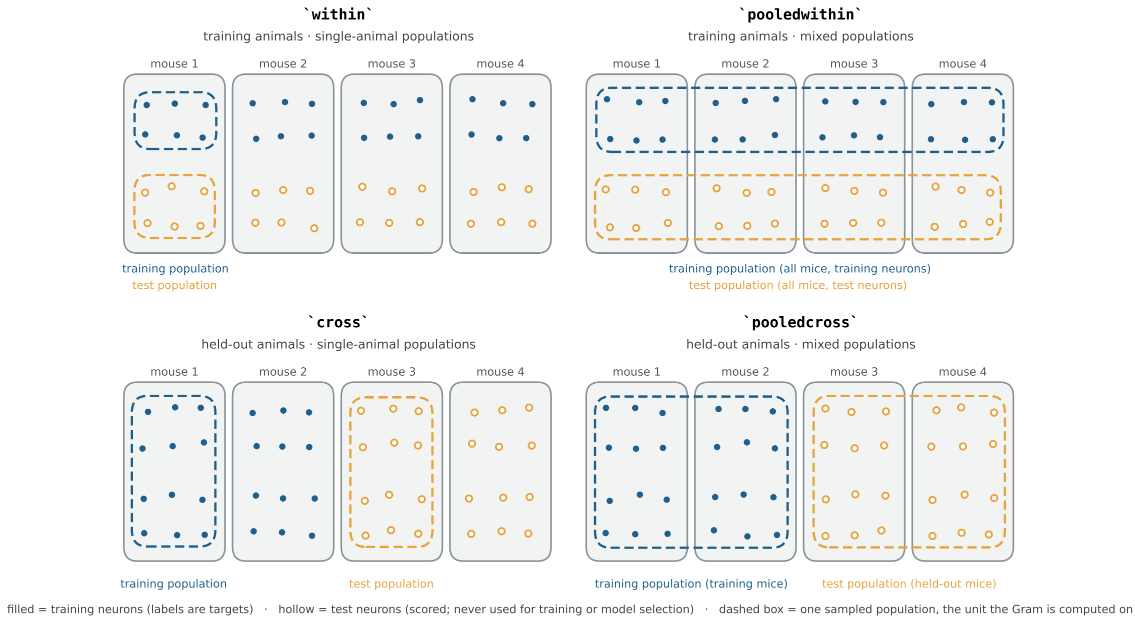

Figure 4: The four Allen decoder regimes. Rows: test neurons from the training animals (each animal's cells halved) or from held-out animals. Columns: each sampled population drawn from one animal or from several. Filled dots are training neurons, hollow dots test neurons, dashed boxes one sampled population each. Training populations come only from training neurons and test populations only from test neurons in every regime.

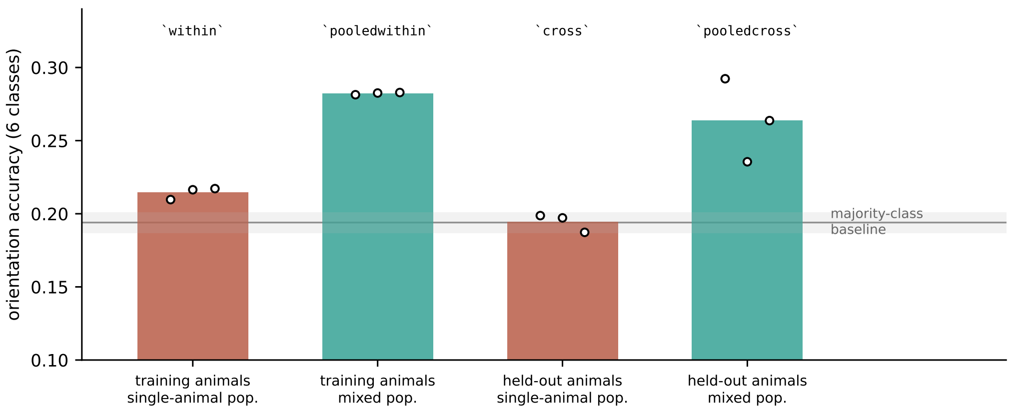

Figure 5: Allen, 33 mice, grating relations: orientation is readable only from populations that mix animals. Bars: mean over splits; points: the three splits. Single-animal populations are at the majority baseline whether the animal was seen in training or not; mixed populations are above it in both rows at the same level.

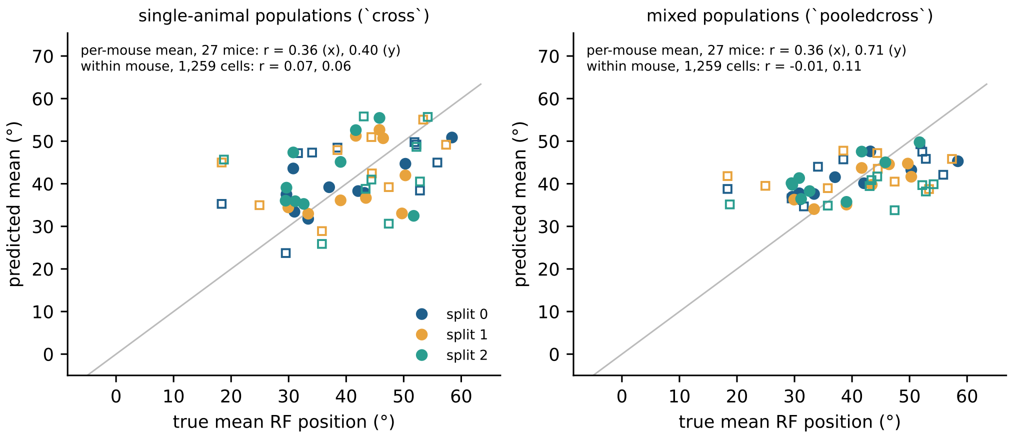

Figure 6: Allen receptive fields: what crosses animals is each animal's screen position. Predicted versus true mean receptive-field position of each held-out mouse (27 points over 3 splits; x squares, y dots), from decoders trained on single-animal (left) or mixed (right) populations. The per-mouse mean is recovered; the position of a neuron relative to its mouse-mates is not.

distance. It is not learned because each mouse contributes 23 to 112 labelled cells. MICrONS, with 11,326 labels in one animal, is where the same decoder reaches neuron-level receptive fields at $R^2 = 0.48$.

3.5 What moves the numbers

Capacity pays where labels are plentiful and stops where they are scarce (Fig. A1). Twin receptive fields go from 0.34 to 0.48 to 0.51 R^2 between 2.2M, 17M and 57M parameters; twin orientation is flat at 0.42 from 7M upward; in-vivo orientation gains one point. Every model above 2M memorises its training neurons within 4 to 17 epochs (training loss 0.001 to 0.06) and is early-stopped on the held-out selection slice (Fig. A4). Population size saturates by 512 neurons in MICrONS and 256 on Allen (Fig. A2). Recomputing training Grams from a random subset of stimulus conditions helps only with capacity, and only where the dimensions are exchangeable conditions: it lifts Allen orientation at 17M and in-vivo receptive fields at 75% of the bins, hurts the 2M model below 75%, and destroys a Gram built on principal components (Fig. A3). On Allen, the 17M augmented model and the 2.2M standard model are equal within seed noise (0.293 ± 0.012 against 0.286 to 0.294), and split-to-split variance exceeds both (Fig. A5).

4 Discussion

The correlation matrix of a few hundred cortical neurons, with the stimulus unknown and no neuron labelled, determines each neuron's preferred orientation to within a few points of what a fully labelled linear readout achieves in vivo, and its receptive-field position to $R^2 = 0.23$ in vivo and 0.48 on the digital twin. The mechanism is the one the symmetry analysis exposes: the relations fix the class structure up to the symmetries of the content, and a small, consistent departure from that symmetry fixes the frame.

Across animals the picture splits by content. Orientation lives in the relations between tuning profiles, which exist between any two neurons that saw the same conditions, in the same brain or not; it does not live in the within-circuit correlations of a single animal at the population sizes available here. Receptive-field position is carried at the level of an animal's patch, and the neuron-level layout needs more labels per animal than these recordings provide. Both statements are about what correlations under a shared stimulus contain, and both are measured under a protocol in which the test neurons, and in the held-out regimes the test animals, never touch training or model selection.

Three limits. MICrONS is one animal, and its training and test halves share recording sessions; the Allen held-out-animal cells are the cleaner generalisation test, and their variance across mouse splits is large. Orientation labels are noisy in both datasets (in vivo and twin labels agree on 70% of MICrONS neurons at 8 classes; static and drifting gratings on 45 to 67% of Allen cells), which caps every labelled and label-free number alike. The twin substrate is a model fitted to the same recordings, so the in-vivo results carry the claim about cortex and the twin results show what cleaner correlations would give.

The symmetry result is an instance of a general expectation: a representation that is equivariant to a group has a group-fixed part, identical across individuals up to relabelling, and an individual remainder [20]. The matching-over-relabellings test is the discrete form of unsupervised alignment of similarity structures across individuals by optimal transport [21]. What this paper adds is the measurement in cortex, at the level of single neurons, and the finding that the remainder, the anisotropy, is what makes the absolute readout possible.

Code and data. TODO: repository URL. All data are public releases.

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A Scaling and controls

Figures A1 to A5 report the sweeps behind Section 3.5: decoder size, population size, relation augmentation, training dynamics and split variance.

B Labelled reference models

TODO: ridge readouts from each neuron's correlations to labelled training neurons (MICrONS in vivo 0.41 / 0.32, twin 0.49 / 0.67; Allen leave-one-mouse-out 0.35 for orientation, 0.20 for receptive fields), with the caveat that ridge is one linear labelled model and the decoder exceeded it on Allen receptive fields.

C Receptive-field decomposition per split

TODO: per-split table of absolute R^2, mouse-level correlation and within-mouse correlation for both regimes.

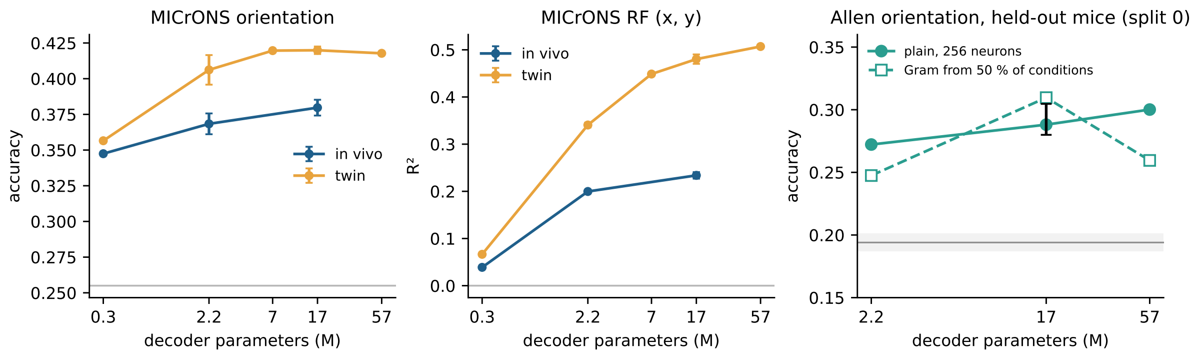

Figure A1: Scaling with decoder size. Test metric versus number of decoder parameters at fixed population size (512 neurons MICrONS, 256 Allen). Error bars: s.d. over 3 seeds where run. On MICrONS receptive fields the gain continues to 57M; on orientation it stops at 7M. On Allen the augmented 17M model equals the 2M standard configuration within seed noise, and augmentation on the 57M model hurts.

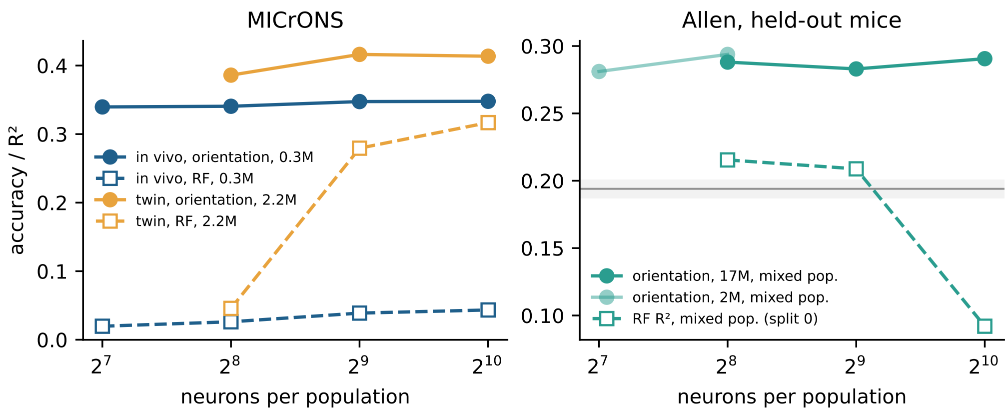

Figure A2: Population size. Test metric versus the number of neurons in each sampled population. Beyond a few hundred neurons the additional companions add redundant context.

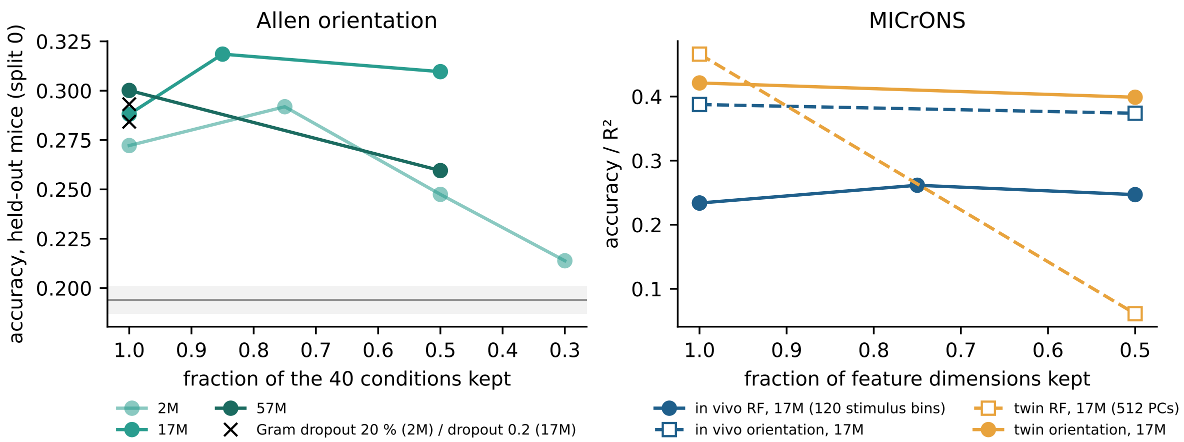

Figure A3: Relation augmentation. Each training population's Gram is recomputed from a random subset of the response dimensions; test Grams use all dimensions. On Allen the augmentation helps only with capacity. On MICrONS it helps only in-vivo receptive fields, costs a point on in-vivo orientation, and destroys the twin Gram, whose dimensions are principal components rather than exchangeable conditions.

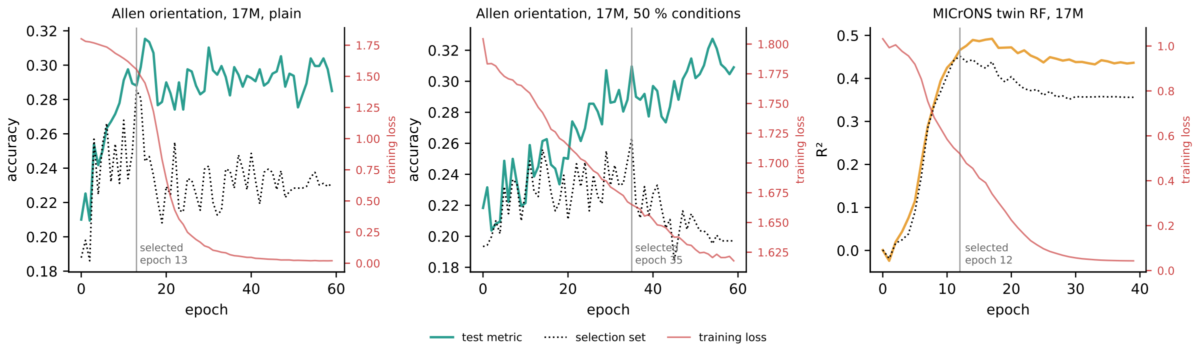

Figure A4: Training dynamics and epoch selection. Test metric (solid), selection-set metric (dotted) and training loss (red, right axis) per epoch. The selection set, never the test set, picks the epoch (grey line). Without augmentation the training loss collapses within about 15 epochs.

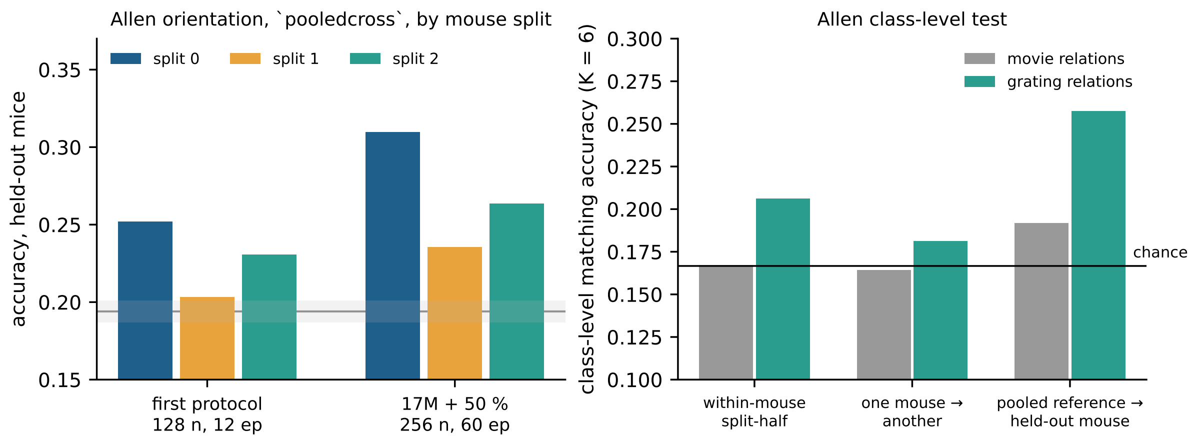

Figure A5: Allen split variance and the class-level test. Left: `pooledcross` orientation accuracy on the same three held-out-mouse splits for the first protocol and the final configuration. Right: the class-level test with relations from natural movies versus drifting gratings; only the grating relations move every test off chance.